FINAL YEAR PROJECT PROPOSAL

KidBuddy

An AI-Powered Safe Learning & Browsing Companion for Children with Neurodevelopmental Conditions

Covering five neurodevelopmental condition domains:

Learning Disabilities • Intellectual Developmental Disabilities • Autism Spectrum Disorder • ADHD • Communication Disorders

Domain: Artificial Intelligence / Machine Learning / Computer Vision / Assistive Technology

1. Executive Summary

KidBuddy is a proposed cross-platform (web + mobile) application designed as a safe, structured, and AI-assisted digital companion for children living with neurodevelopmental conditions. The project covers five condition domains: Learning Disabilities (dyslexia, dysgraphia, and dyscalculia), Intellectual Developmental Disabilities (ID/IDD), Autism Spectrum Disorder (ASD), ADHD, and Communication Disorders (including speech/language delay). The platform is not a diagnostic or clinical tool and is explicitly not designed to replace a therapist, psychologist, special educator, or Speech-Language Pathologist (SLP). Instead, it acts as a home-reinforcement and safe-browsing layer that lets parents and caregivers extend and practise the exercises, instructions, and goals set by the child's therapist, in a controlled, distraction-free, and age-appropriate digital environment.

The system uses Artificial Intelligence, Machine Learning, and Computer Vision to make reinforcement activities adaptive and engaging rather than static worksheets: a read-along module tracks a child's speech against on-screen text; a facial-emotion module teaches emotion recognition to autistic children through their own webcam; a pronunciation/vowel module scores speech-delayed children's speech against reference audio; a life-skills sequencer supports children with intellectual developmental disabilities in building everyday adaptive routines; and adaptive attention games and routine calendars support children with ADHD. Every module is designed around one guiding principle: the therapist gives the plan, KidBuddy helps the family practise it, and reinforces the instructions, rules, and practices given to them.

2. Background and Motivation

2.1 The Problem

Neurodevelopmental conditions such as learning disabilities, intellectual developmental disabilities, ASD, ADHD, and communication disorders affect a significant proportion of children worldwide, and Pakistan is no exception. However, the response to these conditions in Pakistan lags far behind global standards, both socially and technologically.

2.2 Pain Points

- **Low public awareness:** In Pakistan, terms like dyslexia, intellectual disability, ASD, and ADHD are still widely misunderstood, frequently mistaken for laziness, poor upbringing, or slowness, rather than being recognised as neurodevelopmental conditions that require structured support. This delays diagnosis and intervention, often by years, or the condition goes undiagnosed for the person's lifetime.
- **Severe shortage of specialists:** The number of qualified Speech-Language Pathologists, special educators, occupational therapists, and child psychologists in Pakistan is extremely small relative to the population, and they are concentrated almost entirely in a handful of large cities.

- **Cost and accessibility:** Regular therapy sessions are expensive and time-consuming for many families, and children in smaller cities or rural areas often have no access to a specialist at all, meaning progress depends heavily on inconsistent at-home practice with no proper tools to guide it.
- **Stigma:** Fear of social judgement often causes parents to delay seeking help or to hide a diagnosis, which reduces the consistency of the support the child receives at home.
- **Little to no local tech investment:** Unlike regions such as the US and UK, very little assistive/ed-tech software has been built specifically for Pakistani children with neurodevelopmental conditions — both in terms of local language support (Urdu) and awareness-driven design. Most available apps are generic, foreign-made, and English-only.
- **Unsafe, unstructured screen time:** Children with these conditions are especially vulnerable to unstructured internet/device use e.g. algorithm-driven content, ads, and inappropriate material, which can be more destructive given their diagnosis; yet parents are given no safe, purpose-built alternative that is both protective and therapeutically useful.
- **Disconnect between therapy sessions and home practice:** Therapists give instructions (word lists, articulation drills, behavioural routines, life-skills goals) that are usually written on paper and easily forgotten, inconsistently practised, or not practised in an engaging way that holds the child's attention.

These pain points define the core justification of this FYP: to build a locally-relevant, AI-driven, safe digital environment that closes the gap between what a therapist prescribes and what actually happens at home, while contributing meaningfully to an under-served area of Pakistani tech and assistive-computing research.

3. Project Objectives

- Design and develop a safe, closed, ad-free, parent-controlled mobile/web environment for children with neurodevelopmental conditions.
- Build AI/ML-driven reinforcement modules across five condition domains: **Learning Disabilities** (dyslexia, dysgraphia, dyscalculia), **Intellectual Developmental Disabilities** (adaptive life-skills and foundational cognitive support), **Autism Spectrum Disorder** (emotion recognition and communication), **ADHD** (attention and routine), and **Communication Disorders** (pronunciation and language articulation).
- Incorporate Computer Vision as a core pillar of the system — facial landmark and emotion detection, gaze/pointer tracking for reading, mouth/viseme tracking for speech, and handwriting-stroke analysis.
- Provide a parent/guardian dashboard that lets caregivers input the specific instructions or goals given by their child's therapist, so the app's exercises stay aligned with the actual treatment plan.

- Track a child's progress over time (accuracy, streaks, engagement, adaptive-functioning milestones) and surface it to parents in a simple, non-clinical, encouraging format.
- Explicitly position the system as a reinforcement and awareness tool, never as a diagnostic tool or a replacement for professional therapy.

4. Scope

4.1 In Scope

- A child-facing application with five condition-specific module groups: Learning Disabilities, Intellectual Developmental Disabilities, Autism Spectrum Disorder, ADHD, and Communication Disorders.
- A parent/guardian-facing dashboard for setup, therapist-instruction entry, safe-content controls, and progress viewing.
- On-device / server-side AI models for speech recognition & scoring, facial emotion recognition, gaze/pointer tracking, and handwriting analysis.
- A restricted, curated content layer (“safe browsing”) so the child cannot reach anything outside the app's approved activities.

4.2 Out of Scope

- Clinical diagnosis of any condition.
- Replacing therapy sessions, medication management, or professional behavioural intervention plans.
- General-purpose web browsing or unrestricted internet access for the child profile.

5. Users and Scope of Conditions

5.1 Target Users

User	Role in the System
Child (primary user)	Interacts with condition-specific modules through a simplified, distraction-free, gamified UI appropriate to their age and condition.
Parent / Guardian	Sets up the child profile, enters therapist instructions/goals, configures safe-content limits, monitors progress, receives reports.

Therapist / SLP / Special Educator (indirect user)	Does not log into the app directly in the MVP, but their prescribed exercises, word lists, and adaptive-functioning goals are the source of truth that parents configure the app against.

5.2 Neurodevelopmental Conditions Overview

1. Intellectual Developmental Disabilities (ID/IDD)

Affects both intellectual functioning (reasoning, problem-solving, learning) and adaptive functioning (communication, self-care, social skills, managing everyday life). ID/IDD is not simply “a child who learns slowly” — it involves broader, ongoing difficulties across both intellectual and adaptive domains.

Core difficulty: generalising what is learned (intellectual functioning) alongside self-care, communication, and everyday independence (adaptive functioning). Because ID/IDD spans both domains, its module intentionally reuses and re-frames several features from the other four domains rather than duplicating them, and adds a small set of features specific to adaptive-living skills.

2. Autism Spectrum Disorder (ASD)

Typically involves differences or difficulties in social communication and interaction, understanding and using social cues, and restricted or repetitive behaviours/interests; sensory-processing differences can also occur. Autism exists on a spectrum, so children can have very different strengths and support needs.

Core difficulty: recognising emotions and non-verbal cues, expressing needs, social communication.

3. ADHD

Can involve inattention, hyperactivity, impulsivity, and difficulties with executive functions such as organisation, planning, and working memory, with different presentations including predominantly inattentive and predominantly hyperactive/impulsive.

Core difficulty: sustained attention, working memory, routine and time management.

4. Learning Disabilities (Specific Learning Disorders)

Includes dyslexia (reading), dysgraphia (writing/written expression), and dyscalculia (mathematics). These are not forms of intellectual disability — a child can have average or above-average intelligence and still have a significant specific learning disorder.

- Dyslexia — core difficulty: letter/word reversals, slow or inaccurate reading, spelling mistakes.

- Dysgraphia — core difficulty: letter formation, spacing, pencil control.
- Dyscalculia — core difficulty: weak “number sense,” meaning difficulty judging quantity at a glance, mapping numerals to the amounts they represent, placing numbers on a mental number line, and holding math facts in working memory.

5. Communication Disorders

Includes difficulties with language (understanding or producing language), speech sounds (producing sounds clearly), fluency (e.g. stuttering), and social/pragmatic communication (using communication appropriately in social situations). Language delay and speech delay fall under this domain, though “speech delay” itself is a broad descriptive term rather than one specific diagnosis.

Core difficulty: late talking, vowel/consonant articulation, mouth-movement coordination, and, in social/pragmatic cases, using communication appropriately in social situations.

6. Functionalities

6.1 Read-Along Tracker & Struggle-Word Revision

Used by: Dyslexia, Dysgraphia, Communication Disorders

As the child reads a sentence or paragraph aloud, real-time speech recognition follows along with the on-screen text, highlighting the current word (the “pointer”) and flagging any word that's mispronounced or skipped so the child can repeat it. Every flagged word is logged automatically into a personal “words I struggle with” deck, which resurfaces those words for spaced revision — so the difficulty ladder rises and falls based on the child's own mistake history rather than a fixed curriculum.

6.2 Adjustable Reading Assist (Text-to-Speech & Dyslexia-Friendly Formatting)

Used by: Dyslexia, Dysgraphia, Communication Disorders

The child or parent can upload any block of text (a PDF, a photographed page, or typed text) and the app reformats it for easier reading: dyslexia-friendly fonts (e.g. OpenDyslexic), increased letter and line spacing, shorter line lengths, and colour overlays. These visual adjustments reduce letter-crowding and reversal confusion for many children. The same text can also be read aloud via text-to-speech for children who need audio support alongside, or instead of, visual reading.

6.3 Word & Phonics Practice Suite

Used by: Dyslexia, Dysgraphia, Communication Disorders

Combines staged phonics word lists (built from Dolch/Fry sight words) with a colour-coded syllable/phoneme breakdown tool (e.g. “ele-phant,” each syllable a different colour) and an image-word-audio vocabulary builder with spaced repetition. Word-building follows a fixed progression — single sounds → syllables → whole words → short sentences — so the same engine serves a child on early phonics and a child on functional vocabulary. Difficulty and word selection are driven by the same struggle-word history built up in the Read-Along Tracker (6.1), so a word a child gets wrong while reading also surfaces here for targeted practice.

6.4 Multisensory Spelling & Letter-Sound Practice

Used by: Dyslexia, Dysgraphia, Communication Disorders

Combines the Spelling Checker Game — audio plays a word, the child types or selects letters, and mistakes are analysed for dyslexia-typical error patterns (letter reversals such as b/d or p/q, letter-order swaps) rather than simply marked “wrong” — with a letter-sound matching game (the child traces a letter shape while hearing its sound and seeing an image, reinforcing motor, auditory, and visual pathways together in the Orton-Gillingham style), and multisensory spelling practice (a word is spoken aloud, the child types or writes it, and gets combined audio and visual feedback instead of relying on visual memory alone).

6.5 Handwriting & Letter-Formation Suite

Used by: Dysgraphia (also relevant to Dyslexia, given the overlapping letter-reversal pattern)

Guided letter-tracing exercises show animated stroke-order arrows on a touchscreen/stylus, with immediate feedback if the child starts or draws a stroke the wrong way — dysgraphic children often struggle with stroke sequencing specifically, not just the final shape. A CNN trained on handwriting-sample datasets scores handwriting quality and flags reversed or malformed letters. A line-and-space awareness tool highlights when letters drift above, below, or crowd the guide lines. A forgiving handwriting spellcheck recognises common dysgraphic formation errors (reversed letters, inconsistent sizing) in scanned handwritten work, instead of just marking it wrong.

6.6 Vowel & Phoneme Trainer (with Urdu Track)

Used by: Communication Disorders, ADHD, ASD

The child watches a short animation of the correct mouth movement for a target sound, then attempts it themselves. The attempt is scored on two channels. First, the produced audio is compared against reference vowel/phoneme formant data — how the sound should sound. Second, and this is where the camera comes in, the front camera optionally tracks the child's own lip and jaw movement using facial-landmark detection (extracting mouth contour, jaw-opening, and lip-rounding via a lightweight landmark model such as MediaPipe Face Mesh), and compares that movement sequence against a reference viseme pattern for the target sound — how the sound should look being made. This gives the child two separate, understandable feedback signals instead of one opaque pass/fail score, which matters most for sounds that are

easy to mimic by ear but hard to physically form correctly. An Urdu-localised track applies the same audio-plus-viseme pipeline to Urdu vowels and phonemes for Urdu-speaking children.

6.7 Emotion Recognition Trainer

Used by: Autism Spectrum Disorder, ADHD

Uses the child's own webcam feed, or provided images/videos, to run two game modes. In “name that emotion” mode, the system shows a face — a still image, a short clip, or the child's own live expression — and asks the child to identify the emotion. In “mirror the emotion” mode, the app displays a target emotion and asks the child to make that expression themselves; face detection plus facial-landmark extraction (key points around the eyes, brows, and mouth) then reads the child's live expression and compares it to the target. A lightweight classifier — trained on children's facial-expression data rather than adult faces, since expression intensity and features differ by age — scores the match and gives simple, encouraging visual feedback (e.g. filling in a progress ring as the expression gets closer) rather than a blunt right/wrong. The same landmark data lets the system track which specific emotions a child recognises reliably versus struggles with (e.g. strong on “happy”/“sad,” weak on “surprised”/“scared”), so parents — and indirectly, therapists — can see where to focus.

6.8 Number Sense Studio

Used by: Dyscalculia

One adaptive numeracy app, in place of several separate exercises. It targets the two deficits most consistently linked to dyscalculia — recognising quantity at a glance (subitizing) and judging magnitude — then extends into place value, fractions, time, and word problems.

Camera-based object counting. Instead of only dragging on-screen counters, the child can hold up real fingers, blocks, or counters. A hand/object-tracking model (e.g. MediaPipe Hands) reads the quantity and feeds it into whichever game is active — letting the child move between real objects and on-screen numbers, following the concrete → representational → abstract sequence recommended for dyscalculia.

Six mini-games, one engine:

- *Quick Peek* — dots flash for a second; guess the count without counting one-by-one.
- *Number Line Jump* — drag a number to its place on a line.
- *Build-It Blocks* — base-ten blocks for place value and addition/subtraction.
- *Pizza Fractions* — slice a pizza or bar to build a fraction.
- *Word-Problem Scaffolder* — breaks a word problem into pictured steps before the child answers.
- *Clock Detective* — drag clock hands to a target time (unlocked later, since it leans more on sequencing than number sense).

A single knowledge-tracing model tracks mastery across all six and picks what's next — e.g. a child strong at subitizing but weak at magnitude gets steered toward more Number Line Jump, rather than a fixed lesson order. The Word-Problem Scaffolder also adjusts its step count using the same working-memory signal from the ADHD games in 6.8, since word problems lean on working memory too.

Number Writing Check (reuses the dysgraphia CNN). The handwriting-scoring CNN built for the Dysgraphia suite (6.5) is retrained on digit samples to catch dyscalculia-linked writing errors: reversed digits (3, 7, 9 written backwards), 6/9 confusion, and transposed order (writing 21 as 12). The child traces or writes a number on a touchscreen, or scans handwritten work, and gets feedback on the shape itself — because a child who understands "seven" but writes it unrecognisably needs different help than one who doesn't grasp the concept at all.

6.9 AAC-Style Communication Board

Used by: Autism Spectrum Disorder, Intellectual Developmental Disabilities, Communication Disorders

A picture/symbol and gesture-based board (PECS-inspired) that lets a non-verbal or minimally-verbal child tap images, or use simple recognised gestures, to communicate needs to a parent — hungry, tired, want to play. The same board serves all three domains differently: for ASD it supports non-verbal social communication, for IDD it supports functional everyday requesting, and for communication disorders it gives a child with minimal expressive language a working channel while speech/language skills are built up elsewhere in the app.

6.10 Social-Story Interactive Scenarios

Used by: ADHD, Autism Spectrum Disorder, Intellectual Developmental Disabilities

Short interactive scenarios that model an expected behaviour or response — greeting someone, asking for help, waiting a turn, transitioning between tasks — where the child chooses or acts out a response before seeing the outcome. One engine covers social-communication practice (ASD), self-regulation situations (ADHD), and everyday behaviour modelling (IDD), with the specific scenario library adapted per domain.

6.11 Visual Routines, Life-Skills & Attention Engine

Used by: Intellectual Developmental Disabilities (primary), ADHD (routine & attention components), ASD

One shared, step-by-step visual engine powering four related tools:

- **Self-Care Visual Sequencer —** step-by-step visual + audio sequences for everyday self-care tasks (brushing teeth, handwashing, getting dressed, packing a school bag), checked off one picture at a time. This is the module's core original contribution, targeting adaptive functioning directly.

- **Routine Calendar —** a visual daily/weekly schedule (morning routine, homework, therapy exercises, chores) with reminders and reward stickers/points for completion. A simplified, lower-density variant (fewer items per day, larger icons, stronger repetition) is used for IDD to build daily-living independence; the full-density version is used for ADHD, for time-management.
- **Foundational Concept & Reasoning Game —** matching, sorting, and categorising activities (by colour, shape, size, or category) with adjustable cognitive load, building the basic reasoning and generalisation skills intellectual functioning relies on.
- **Attention & Focus Mini-Games —** short, game-based attention tasks (spot-the-difference, sustained visual search, go/no-go style games) with adaptive difficulty, for ADHD.

Progress on the IDD-facing tools is deliberately framed around adaptive-functioning milestones (e.g. “completed morning routine independently 4/5 days”) rather than accuracy percentages, since the goal is everyday independence, not test-style scoring.

7. Existing sites

7.1 Read-Along Tracker & Struggle-Word Revision

- Read Along by Google – https://readalong.google.com/ (real-time speech tracking + word highlighting)
- RallyReader – https://cuberto.com/projects/rallyreader/
- *Gap:* no known app pairs this with an auto-generated spaced-repetition "struggle word" deck — that combination isn't found elsewhere.

7.2 Adjustable Reading Assist (TTS & Dyslexia Formatting)

- Dyslexia Reader – https://dyslexiareader.com/ (closest full match: PDF/photo upload, dyslexia-friendly reformatting, syllable/onset-rime breakdown, TTS)
- Helperbird / Immersive Reader – https://www.helperbird.com/features/immersive-reader-for-chrome/
- OpenDyslexic – https://opendyslexic.org/

7.3 Word & Phonics Practice Suite

- Nessy – https://www.nessy.com/
- U-M Dyslexia Help resource – https://dyslexiahelp.umich.edu/?p=4453

7.4 Multisensory Spelling & Letter-Sound Practice

- OgStar Reading – https://www.ogstarreading.com/ (full Orton-Gillingham app: sound/symbol, tracing, handwriting + speech recognition feedback)

- Dyscover (Purkayastha et al., IEEE 2012) – academic OG-inspired multisensory prototype
- Writing Wizard (via Perkins) – https://www.perkins.org/resource/using-writing-wizard-app-students-who-are-low-vision-and-have-cvi/
- *Gap:* no consumer app specifically analyses errors for dyslexia-typical patterns (reversals, order-swaps) rather than flat right/wrong.

7.5 Handwriting & Letter-Formation Suite

- Kaligo – https://www.kaligo-apps.com/ (AI scores stroke shape/direction/order in real time, incl. numbers)
- LetterSchool – https://www.letterschool.com/
- Dynamilis/Tegami project (EPFL) – https://www.epfl.ch/labs/chili/Dynamilis (CNN-based dysgraphia detection research)

7.6 Vowel & Phoneme Trainer (Urdu Track)

- Speech Blubs – https://speechblubs.com/ (video-modeling of mouth movements, AR face filters — closest analog, but not scored)
- Otsimo Speech Therapy – voice + visual mouth-movement prompts
- Bolo Tech (Pakistani Urdu speech-therapy platform) – covered here: https://propakistani.pk/2017/02/10/pakistani-duo-made-urdu-speech-therapy-platform-kids/
- *Gap:* no system found doing live camera lip/jaw-landmark tracking scored against a reference viseme pattern, in English or Urdu — this dual audio+viseme scoring appears novel.

7.7 Emotion Recognition Trainer

- Emotion Mirror (Deriso et al., ECCV Workshop 2012) – real-time webcam expression-mirroring research system, closest match
- Emoface – https://www.emoface.fr/en/emoface-play-and-learn-emotions-en/
- Autism: Matching Emotions (App Store)
- *Gap:* none confirmed to use a classifier trained on children's (vs. adult) facial data with per-emotion mastery tracking.

7.8 Number Sense Studio

- Monster Math – https://www.monstermath.app/
- The Number Race (Wilson, Cohen, Dehaene et al.) – academic adaptive dyscalculia software: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1523349/
- Little Digits (App Store) – touchscreen finger-counting, closest to camera-based counting idea (not camera-based)

- *Gap:* no product combines real-object camera counting + six linked mini-games + one knowledge-tracing model; closest precedent is a research-only webcam finger-counter (Crampton & Betke, Boston University).

7.9 AAC-Style Communication Board

- Proloquo2Go – https://www.assistiveware.com/products/proloquo2go
- CoughDrop – https://www.coughdrop.com/
- My Avaaz (Urdu AAC, Pakistan) – https://www.myavaaz.com/

7.10 Social-Story Interactive Scenarios

- Social Story Creator & Library (Touch Autism) – http://touchautism.com/app/social-stories-creator-library/
- GrowTale – https://www.growtale.org/
- *Gap:* existing apps present fixed narrated/personalized stories; none branch dynamically based on the child choosing a response before seeing the outcome.

7.11 Visual Routines, Life-Skills & Attention Engine

- Choiceworks – https://apps.apple.com/app/choiceworks/id486210964 (schedules, self-care steps, rewards)
- FocusPlay – attention/inhibition mini-games (Stop/Go, Stroop-style)
- EndeavorRx – https://www.endeavorrx.com (FDA-authorized adaptive attention game)
- *Gap:* no single product bundles self-care sequencing + routine calendar + reasoning/sorting games + attention games under one adaptive engine.

8. Comparative Analysis of Existing Systems

While each module in KidBuddy has real-world precedents, existing tools are single-purpose products built in isolation from one another — a family managing overlapping conditions must juggle multiple disconnected apps, each with its own account, curriculum, and reward system that cannot be aligned to a therapist's actual plan. Table 8.1 summarises, for each functionality, what current systems already achieve and where KidBuddy diverges.

Table 8.1 — Feature-Level Comparison

Module	Representative Existing System(s)	Existing Capability	KidBuddy's Differentiation

6.1 Read-Along Tracker	Read Along (Google), RallyReader	Real-time speech tracking; highlights word being read	Same live-tracking core, but passages are generated around the child's own family names/hobbies; flagged words feed directly into 6.3/6.4 instead of staying isolated
6.2 Reading Assist	Dyslexia Reader, Helperbird	Reformats uploaded text; dyslexia fonts; TTS	Automatically emphasises the child's *own* struggle words within any uploaded material, not just generic formatting
6.3 Phonics Suite	Nessy	Strong, systematic phonics curriculum	Same rigor, but word order/selection is re-weighted live from the child's mistake history and the therapist's current targets, rather than one fixed sequence for all users
6.4 Multisensory Spelling	OgStar Reading, Dyscover	Orton-Gillingham method; audio + tracing	Classifies *type* of error (reversal, order-swap) instead of flat right/wrong; word bank drawn from the child's actual mistakes
6.5 Handwriting Suite	Kaligo, LetterSchool	AI stroke-order/shape scoring	Tracing of letters,strokes, words tracing (struggled history),numbers etc
6.6 Vowel/Phoneme Trainer	Speech Blubs, Otsimo, Bolo Tech (Urdu)	Video-modeling of mouth movement; audio prompts	Dual-channel scoring — audio-vs-reference *and* live camera lip/jaw tracking vs. a reference viseme, a mouth showing how to pronounce —

			plus a genuine Urdu track, not a separate single-purpose tool
6.7 Emotion Recognition	Emoface, Autism: Matching Emotions	Static photo/video emotion *recognition*	Live webcam emotion *production* practice, scored against a child-trained classifier, with per-emotion mastery tracking, showing pictures and cards of emotions and scenarios and then testing them
6.8 Number Sense Studio	Monster Math	Adaptive numeracy mini-games	Adds camera-based real-object counting and reuses the 6.5 handwriting CNN for digit-reversal checks, rather than three unconnected tools
6.9 AAC Board	Proloquo2Go, CoughDrop, My Avaaz	Picture/symbol communication; My Avaaz is Urdu-only	One board serving three domains, natively bilingual, personalised with the child's own people/objects as icons
6.10 Social Stories	Touch Autism, GrowTale	Pre-made or manually personalised narrated stories	Auto-personalised from a one-time family profile; choice-driven branching (child picks a response, sees the outcome) instead of a fixed script
6.11 Visual Routines & Attention	Choiceworks, FocusPlay, EndeavorRx	Each covers one piece — schedules *or* attention games — separately	One shared visual engine so routines, reasoning games, and attention games behave consistently; progress framed

			as adaptive-functioning milestones, not raw accuracy

The pattern that emerges is consistent: no existing system treats the child as a single profile whose progress, mistakes, and content preferences travel *between* modules and *across* diagnoses.

9. Unique Selling Points (USPs)

9.1 Unified platform across diagnoses, not a fragmented app stack
One login, one child profile, one reward economy spanning all eleven modules. A badge earned in handwriting and a sticker earned on the AAC board sit in the same collection — competitors each run their own separate ecosystem.

9.2 Therapist-configurable, not a fixed curriculum
Every competitor ships a built-in lesson order. KidBuddy lets the child's actual SLP/OT/special educator (or a parent relaying instructions) enter that week's targets — specific sounds, sight words, routine steps — and every relevant module re-weights around them. This makes KidBuddy a genuine extension of clinical practice rather than a generic gamified workbook running in parallel.

9.3 Deep, profile-driven personalisation
A one-time "About My Child" profile (family and pet names, favourite sport or show, hobbies, hometown) auto-populates content across the platform — read-along passages, phonics sentences, AAC icons, social stories, even word problems (*"Ali has 3 cricket balls and gives one to his sister Zara…"*). Existing tools either use generic stock content (Nessy, Speech Blubs) or require the parent to manually rebuild personalised material each time (social story apps). No competitor auto-generates personalised content platform-wide from a single profile.

9.4 Cross-module memory
A word missed in the Read-Along Tracker automatically resurfaces in Phonics and Spelling. A working-memory signal from the ADHD attention games adjusts how many steps the Number Sense word-problem scaffolder provides. This data-sharing between modules has no equivalent in the market, since competing tools are separate products with no shared child model.

9.5 Native Urdu support, built in rather than bolted on
Urdu-capable tools exist in isolated pockets (Bolo Tech for speech, My Avaaz for AAC) but nothing offers Urdu phonics, Urdu social scenarios, Urdu AAC, and Urdu phoneme training as one coherent, localised experience for Pakistani children.

9.6 Continuity through a single evolving companion
Rather than a different mascot per module (as in every competitor surveyed), one companion character persists across all eleven modules, levels up, and is customised by the child —

reinforcing engagement and making the whole platform feel like one experience rather than eleven separate apps.